

When Adaptation Hurts

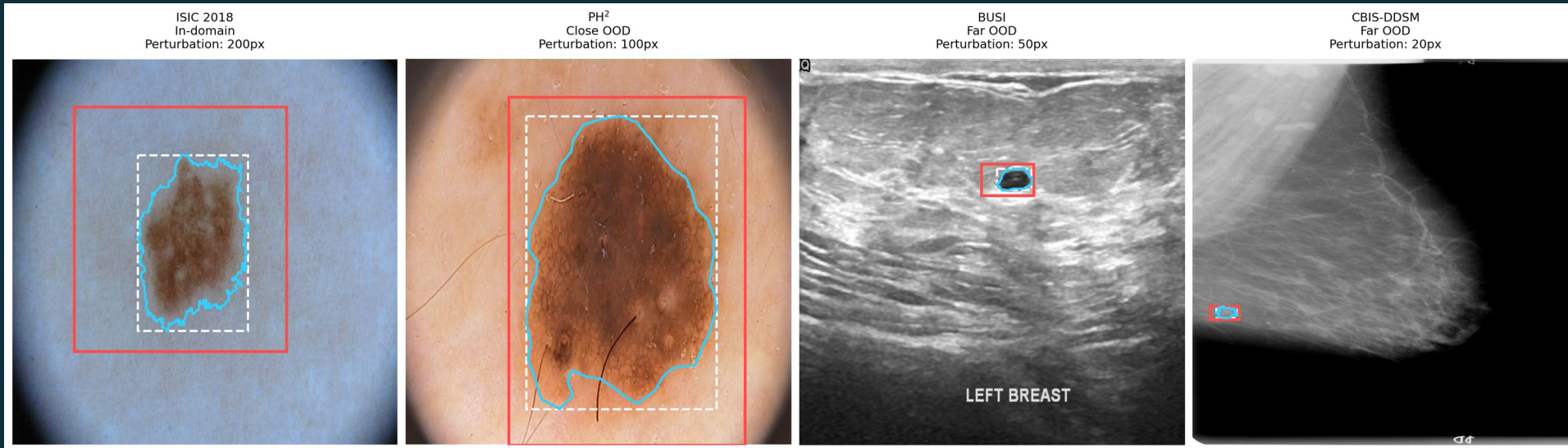
Connecting Representational Drift to OOD Failures in MedSAM Fine-Tuning



github.com/ImSounic/medsam-vpt

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*Same task, four domains,
one loose box.*

Three challenges that are usually studied one at a time

1 ROI sensitivity

MedSAM needs a bounding box. Clinical boxes are loose, inconsistent or missing, and Dice follows box quality.

2 Distribution shift

Change the scanner, organ or imaging modality and accuracy falls. Far-OOD transfer is where foundation models are supposed to earn their keep.

3 Adaptation cost

Fine-tuning helps in-domain but is expensive. LoRA and VPT are cheap, yet sensitive to the data and prompts they are adapted with.

Prior work refines prompts, adapts to a target domain, or detects OOD inputs, each in isolation. **This work** evaluates adaptation strategy × prompt noise × domain shift jointly, then asks which internal representations explain who survives the shift.

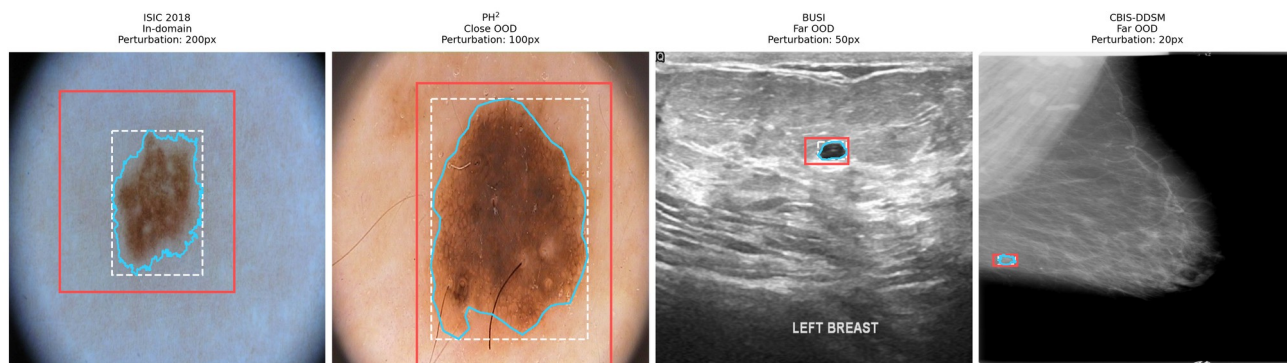


Fig. 1. Tight box (white, dashed), perturbed prompt (red), predicted mask (cyan). The same pixel jitter is a far larger relative change on the small far-OOD lesions in ultrasound and mammography.

Six adapters, one training domain, a drift ladder

Method	Params	What is updated
Zero-shot MedSAM	0	reference
Decoder-only FT	4.1M	mask decoder
VPT-shallow	4.1M	10 prompt tokens at encoder input + decoder
VPT-deep	4.2M	10 prompt tokens at every block + decoder
LoRA ($r = 8, \alpha = 16$)	4.4M	encoder + decoder adapters
Encoder-only LoRA	4.1M	encoder adapters, decoder frozen
Full FT	93.7M	image encoder + mask decoder

Train on ISIC 2018 dermoscopy: 2,594 images, 80/10/10 split, 5 epochs, AdamW, Dice + CE, prompt encoder frozen, 3 seeds. Full FT: LR $1e-5$, batch 1. PEFT: LR $1e-4$, batch 4. Hyperparameters fixed across datasets; 8 GPUs with 16–22 GB, about 8 hours per training.

Evaluate along a drift ladder

ISIC 2018 in-domain · dermoscopy 1,000 test images



PH² close-OOD · dermoscopy 200 images



BUSI far-OOD · breast ultrasound 647 images



CBIS-DDSM far-OOD · mammography 362 images

Prompt protocol. Train with clean boxes, fixed 20 px, or random 0–100 px jitter. Evaluate at 0 / 20 / 50 / 100 / 200 px on 1024^2 images.

+280%

mean box area at 200 px jitter, all datasets

+1,328%

on small CBIS-DDSM lesions: far-OOD is harder by construction

Adaptation helps in-domain and often hurts far-OOD

Dice with clean boxes, models trained with clean boxes, mean of 3 seeds

Method	ISIC (ID)	PH ² (close-OOD)	BUSI (far-OOD)	CBIS-DDSM (far-OOD)
Zero-shot	0.907	0.905	0.823	0.692
Decoder-only	0.949	0.947	0.894	0.828
VPT-shallow	0.945	0.943	0.792	0.566
VPT-deep	0.947	0.946	0.802	0.573
LoRA	0.956	0.957	0.780	0.509
Encoder-only LoRA	0.958	0.960	0.908	0.793
Full FT	0.961	0.958	0.901	0.829

- Every adapter beats zero-shot in-domain. LoRA and both VPT variants **fall below zero-shot** on ultrasound and mammography.
- **Full FT** is the best overall trade-off. **Encoder-only LoRA** is the strongest parameter-efficient method: same adapter as LoRA, decoder left alone.

0.51 vs 0.69

LoRA vs zero-shot Dice on mammography. The best in-domain PEFT adapter is worse than doing nothing far-OOD.

+0.28 Dice

Encoder-only LoRA over standard LoRA on CBIS-DDSM. The only change: the decoder is frozen.

4.1M ≈ 93.7M

Encoder-only LoRA and decoder-only FT stay within 0.04 Dice of Full FT on both far-OOD sets.

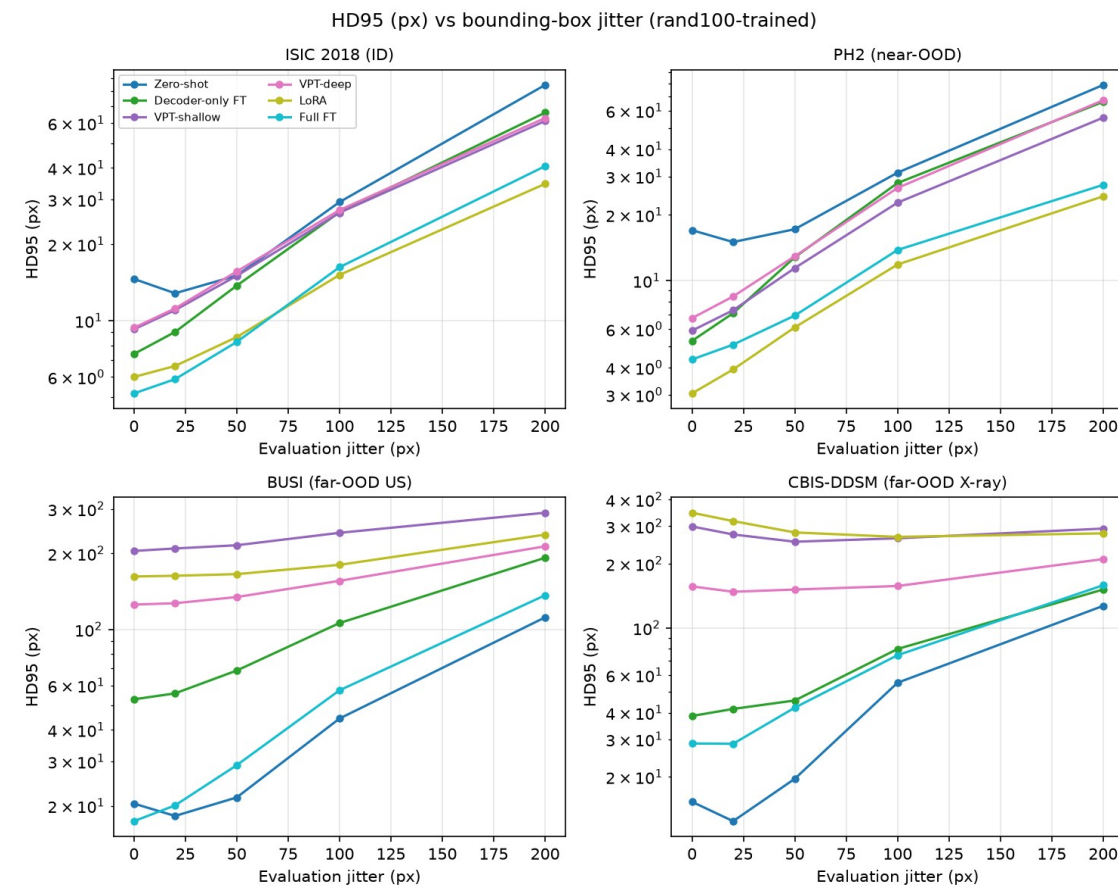
Boundary error shows collapse, not a small overlap loss

HD95 with clean boxes, median [IQR] in pixels at 1024^2 . In-domain and close-OOD medians are 0–7 px for every method.

Method	BUSI (far-OOD)	CBIS-DDSM (far-OOD)
Zero-shot	14 [9, 25]	12 [7, 17]
Decoder-only	4 [2, 17]	4 [3, 6]
VPT-shallow	21 [5, 387]	14 [8, 37]
VPT-deep	25 [6, 394]	18 [10, 359]
LoRA	26 [6, 402]	47 [13, 438]
Encoder-only LoRA	3 [1, 11]	5 [3, 7]
Full FT	4 [1, 13]	4 [3, 6]

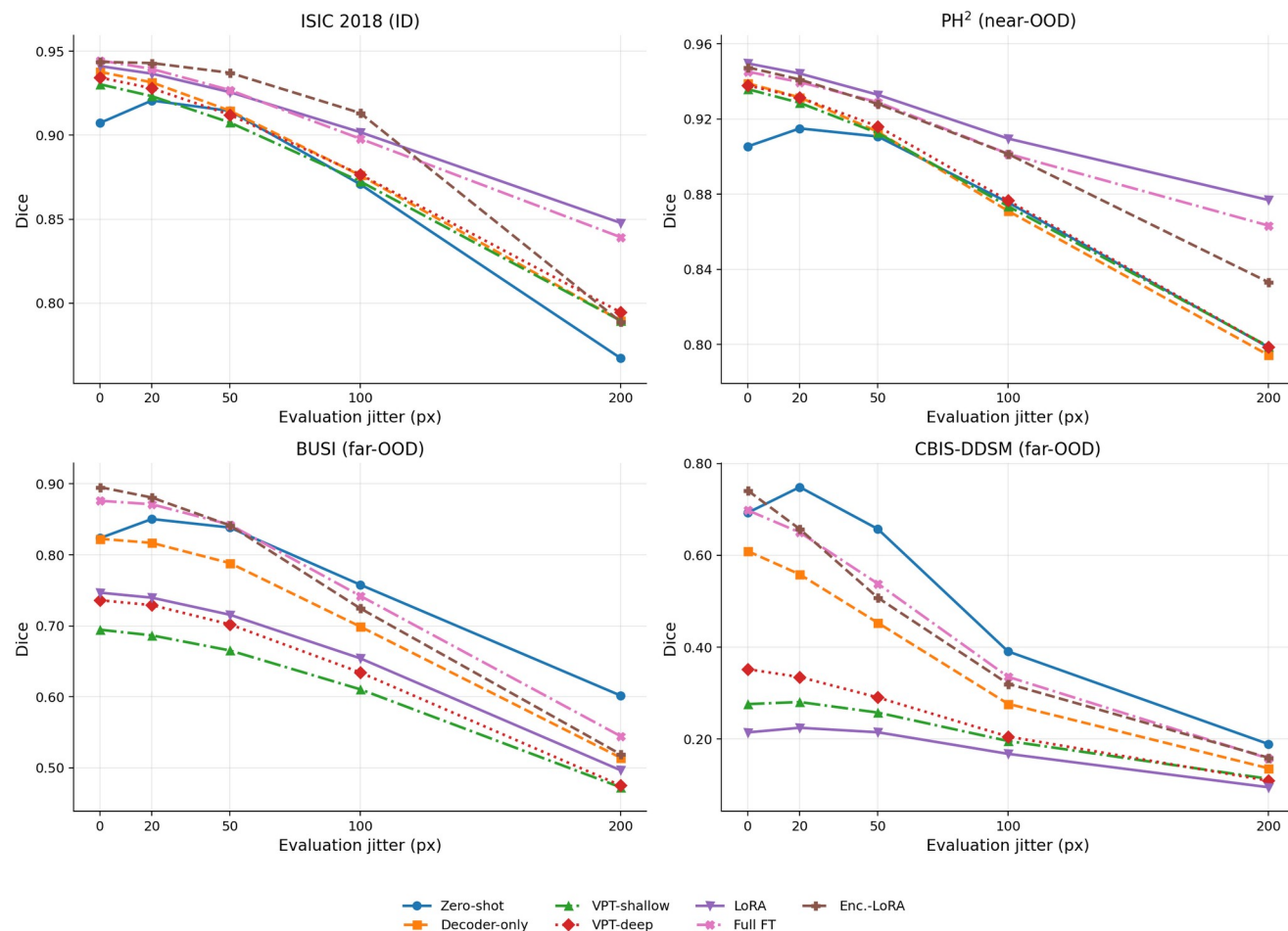
~400 px

Upper quartile of LoRA and VPT boundary error on far-OOD data: a quarter of cases segment the wrong structure entirely. Dice gap to Full FT ~0.12; boundary gap more than tenfold.



HD95 (log scale) vs evaluation jitter, rand100-trained models. LoRA and VPT sit an order of magnitude above the rest on both far-OOD sets at every jitter level.

Clean-box training makes adapters brittle to prompt noise



Dice vs evaluation jitter, averaged over the three training-jitter regimes (paper Fig. 2).

Dice under evaluation jitter, models trained with clean boxes

Setting	Zero-shot	Dec-only	LoRA	Enc-LoRA	Full FT
CBIS, 0 px	0.692	0.828	0.509	0.793	0.829
CBIS, 20 px	0.748	0.706	0.478	0.753	0.744
CBIS, 100 px	0.391	0.303	0.270	0.367	0.336
BUSI, 100 px	0.758	0.659	0.619	0.701	0.672
ISIC, 200 px	0.767	0.710	0.787	0.645	0.712

- Zero-shot **improves** with 20 px of slack and is the best model far-OOD once boxes are 100 px off. MedSAM was pretrained on imperfect boxes.
- Clean-box adapters **unlearn that tolerance**. Even encoder-only LoRA becomes the most jitter-sensitive model in-domain at 200 px.

RESULT 3

Random 0–100 px jitter training gives the most reliable adapters

Mean Dice gain over zero-shot: trained with clean boxes → trained with random 0–100 px jitter (paper Table 1)

Method	In-domain	Far-OOD	Overall
Full FT	−0.006 → +0.034	−0.014 → −0.029	−0.009 → +0.013
Encoder-only LoRA	−0.027 → +0.053	−0.024 → −0.054	−0.023 → +0.007
LoRA	+0.012 → +0.035	−0.142 → −0.228	−0.038 → −0.051
Decoder-only	−0.016 → +0.014	−0.036 → −0.088	−0.023 → −0.022
VPT-shallow	−0.008 → +0.009	−0.127 → −0.230	−0.048 → −0.071
VPT-deep	−0.010 → +0.013	−0.126 → −0.198	−0.049 → −0.058

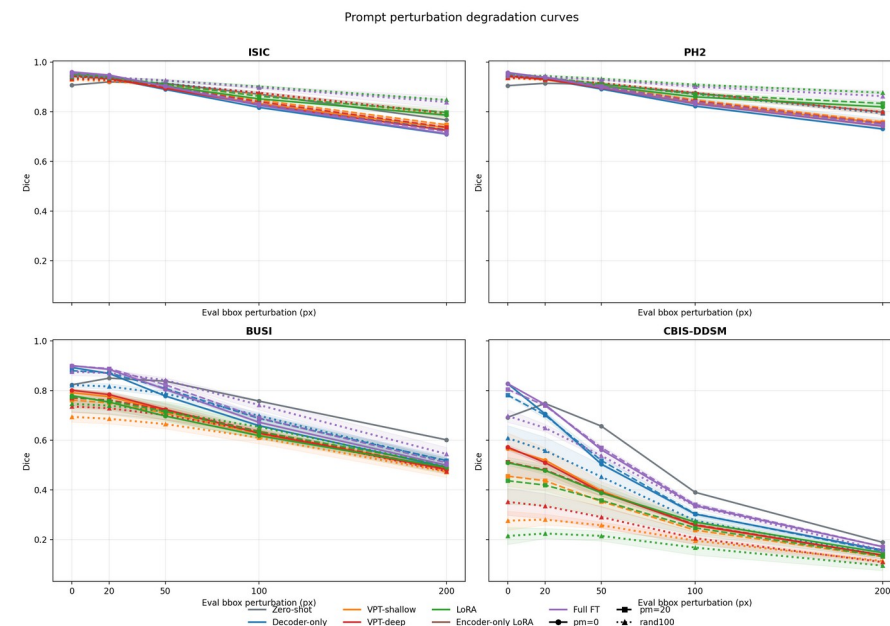
- Jitter training restores in-domain and close-OOD robustness for every adapter. Fixed 20 px training overfits one noise level.
- It costs far-OOD Dice: **small** for Full FT and encoder-only LoRA, **large** for LoRA, VPT and decoder-only.
- Only **Full FT and encoder-only LoRA** end up above zero-shot overall.

0.71 → 0.84

Full FT, ISIC at 200 px jitter,
clean → rand100

0.51 → 0.21

LoRA, CBIS-DDSM clean boxes,
clean → rand100



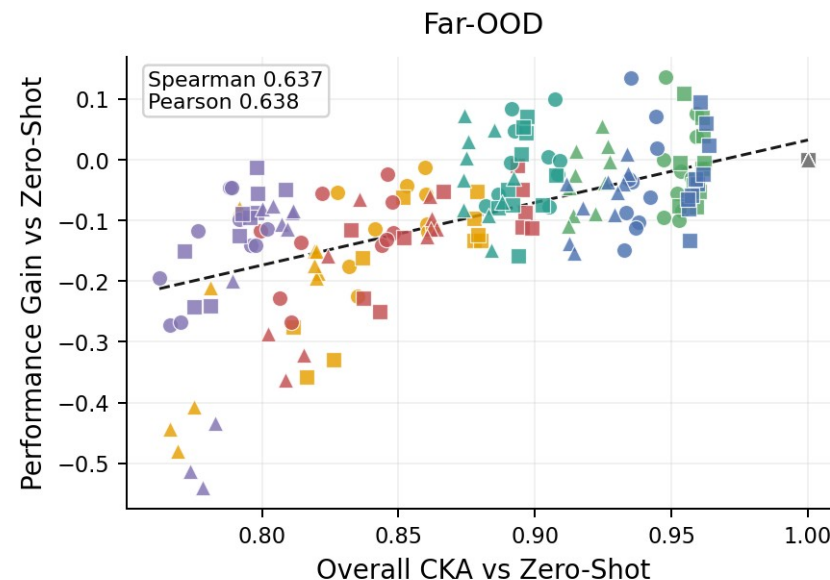
Solid: clean-trained · dashed: fixed 20 px · dotted: random 0–100 px. Dotted curves are flattest in-domain and lowest on CBIS-DDSM.

Far-OOD loss tracks decoder drift, not encoder drift

Spearman ρ between linear CKA (adapted vs zero-shot MedSAM) and Dice gain over zero-shot. $n = 3,000$ pooled observations; FDR-corrected * $q < 0.05$, *** $q < 0.001$

CKA feature	ID (ISIC)	Close-OOD (PH ²)	Far-OOD (BUSI / CBIS)
IoU token layer	0.10	-0.02	0.76 ***
Output layer	0.06	0.00	0.76 ***
Decoder layers	-0.08	-0.02	0.73 ***
Upscaled embedding	0.20	0.29 *	0.71 ***
Encoder layers	-0.28 *	-0.28 *	0.09

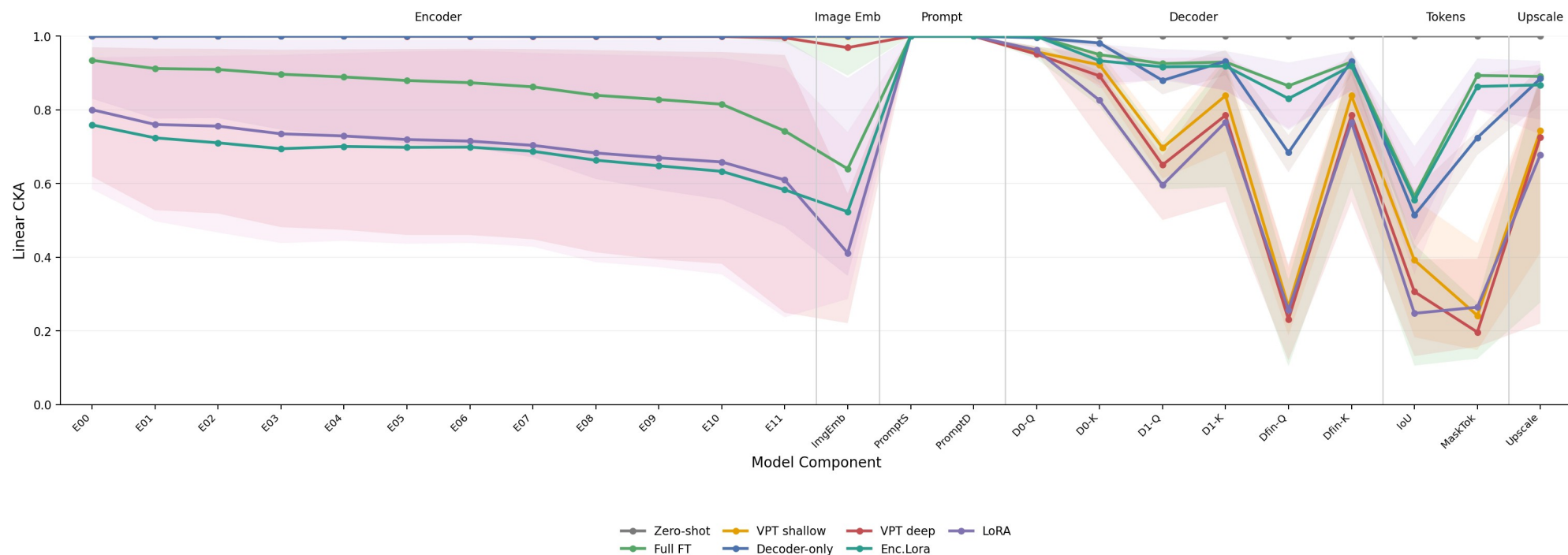
- **Decoder-side similarity predicts far-OOD gain; encoder similarity does not**, and is weakly negative in-domain.
- Decoder CKA on CBIS-DDSM follows the Dice ranking exactly: Full FT 0.95 · decoder-only 0.94 · encoder-only LoRA 0.91 · VPT-shallow 0.83 · VPT-deep 0.80 · LoRA 0.76.
- **Encoder-only LoRA works** because it adapts the encoder to the visual shift while preserving the decoder pathway.



Far-OOD Dice gain vs overall CKA, one point per method × training jitter × evaluation jitter (Spearman 0.64). Bottom-left cluster: LoRA and VPT. Top-right, next to zero-shot: Full FT and encoder-only LoRA.

Caveat. Pooled observations share images and backbones, so the effective sample size is well below $n = 3,000$. Treat the p-values as descriptive; the 0.7 vs 0.1 contrast is the evidence.

Where the representation drifts: component-wise CKA to zero-shot MedSAM



Encoder (E00-E11, image embedding)

LoRA, VPT and encoder-only LoRA all drift, down to 0.4–0.6 at the image embedding. Encoder-only LoRA drifts as much as LoRA and is still robust, so encoder drift is not the problem.

Decoder attention, IoU and mask tokens

LoRA and VPT collapse to 0.2–0.3 at the final attention queries and the IoU and mask tokens. Decoder-only, Full FT and encoder-only LoRA stay at 0.7–0.9.

What the decoder does

It turns a prompt into a mask. That mapping is the transferable part of MedSAM; adapters that rewrite it throw the far-OOD ability away.

Which adapter, which training regime

Compute available, far-OOD data expected	Full fine-tuning at LR 1e-5	Best trade-off in every regime; 93.7M parameters.
Parameter-efficient adaptation	Encoder-only LoRA	Best PEFT far-OOD, competitive with Full FT, 4.1M parameters. Keep the decoder frozen.
Only the target domain, clean prompts	Standard LoRA	Largest in-domain and close-OOD gains; do not deploy it far-OOD.
Any regime	Train with random 0–100 px box jitter	Fixed-jitter training overfits one noise level.
Any regime	Avoid VPT on MedSAM	Negligible in-domain gain, large far-OOD loss.
Severe prompt noise on far-OOD data	Zero-shot MedSAM as fallback	Still hard to beat once boxes are 100 px off.

Read the ranking, not the absolute Dice

1 Asymmetric ladder

One training domain (dermoscopy) and two far-OOD targets. Modality shift is confounded with lesion-type mismatch; the absolute far-OOD Dice is a robustness indicator, not a clinical estimate.

2 Jitter difficulty is dataset-dependent

200 px is +152% box area on ISIC but +1,328% on CBIS-DDSM. Far-OOD evaluation is harder by construction.

3 CKA correlations are descriptive

Observations share images and backbones across methods, seeds and jitter levels, so the effective sample size is well below $n = 3,000$.

4 Fixed recipe

5 epochs, one hyperparameter set for all datasets, LoRA rank 8 with $\alpha = 16$, 10 prompt tokens per insertion point. VPT is an additive perturbation at fixed positions because SAM's windowed attention prevents token prepending.

Three things to remember

1

Adaptation can hurt.

LoRA and VPT beat zero-shot in-domain and fall below it on far-OOD modalities, with boundary errors in the hundreds of pixels.

2

Prompt noise must be trained for.

Random 0–100 px jitter gives the most reliable adapters; only Full FT and encoder-only LoRA end up above zero-shot overall.

3

The decoder is the locus.

Preserving decoder and output representations predicts far-OOD robustness ($\rho \approx 0.73\text{--}0.76$); encoder similarity does not ($\rho = 0.09$). Encoder-only LoRA is the CKA-informed adapter.



Code, configs, checkpoints
github.com/ImSounic/medsam-vpt

Thank you. Questions? · Poster session 12:30–13:30 today